15 DECEMBER 2020

APS & APS-U UPDATES

DENNIS MILLS

DEPUTY DIRECTOR, ADVANCED PHOTON SOURCE

LS-CAT BOARD MEETING

APS OPERATIONS STATUS

- The APS/ANL is in Limited Operations mode - anticipating to remain in Limited Operations mode until at least the end of the calendar year.
- With Limited Ops, APS has ~190-235 staff onsite at any given time, in direct support of APS-U and APS Operations work scopes.
- Majority of beamlines XSD beamlines have now have some sort of remote access mode available to users.
- 2020-3 User Run started on October 1, 08:00

– Total Amount of User Time	1409.0 Hours
– Delivered Beam	1337.2 Hours
– Percentage of Scheduled Time	94.9 %
– Mean Time Between Faults (MTBF)	74.3 Hours

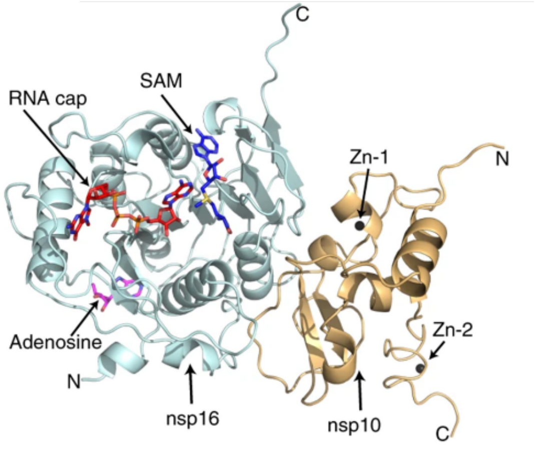
- Uncharacteristically large number of faults; sign of the aging machine?
- Reliability has improved in the second part of the run, inching up to 95%

GENERAL USER PROGRAM UPDATE

- Access:
  - Remote/mail-in experiments remain priority. ANL users from other ANL directorates permitted onsite for experiments, capped at no more than 2 users per ESAF.
    - Requires ESAF review with DOE Argonne Site Office.
    - Due to review cycle, requiring that ESAFs for Argonne employee users onsite submitted 3-4 weeks ahead of experiment start.
  - External (non-Argonne) users permitted on case-by-case basis after ALD approval of submitted proposals/ESAFs
    - APS received 12 outside (non-ANL) user proposals and approved/accepted 7 of those proposals.
- For 2021-1 cycle, APS received 580 new proposals. (That’s an increase over the 425 new proposals we received for the 2020-3 cycle.)
- The Beamline Allocation Committee (BAC) met Monday, Dec 7, allocating time for the 2021-1 cycle, leaving the last 3 weeks in April unallocated to be filled at the discretion of the beamline scientists.

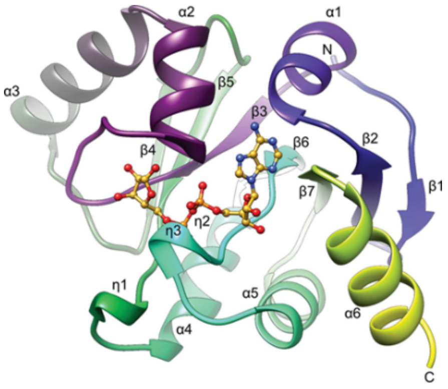
SARS COV-2 RESEARCH AT THE APS

- APS ran an additional 500+hrs over the summer specifically to support COVID-19 research by academics and pharma.
- Data collected at APS beamlines have to date resulted in over **100 structures** deposited in the Protein Data Base.
- Since the start of the pandemic **71** user groups, and **285** unique users have used **20** beamlines for a total of **8172** hours of beam time for COVID-19-related research.
- Much of our basic understanding of the COVID-19 virus is due to work performed at the APS.
- APS also supported additional research related to the study new filter materials for N95 masks



Structure of SARS-CoV-2 nsp16/nsp10 complex with RNA cap (red stick), and S-adenosyl methionine or SAM (blue stick)

"Structural basis of RNA cap modification by SARS-CoV-2," Viswanathan et al., *Nature Communications* (https://doi.org/10.1038/s41467-020-17496-8)



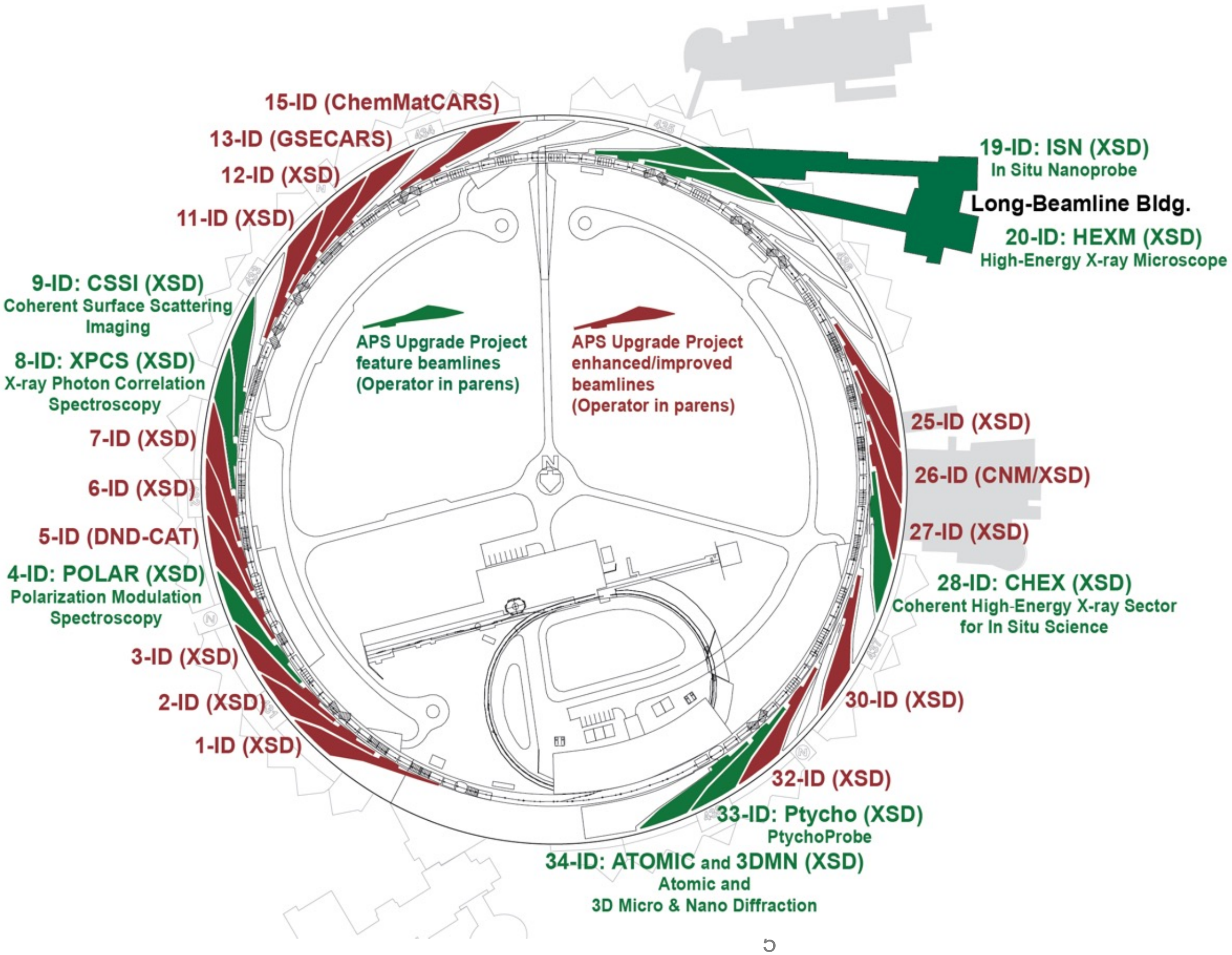
Structure of SARS-CoV-2 nsp3 ADRP. Ribbon diagram shows ADRP-APO2. ADPr ligand molecule is shown based on superposition with the ADRP–ADPr structure

"Crystal structures of SARS-CoV-2 ADP-ribose phosphatase: from the apo form to ligand complexes," Michalska et al, *International Union of Crystallography Journal* (https://doi.org/10.1107/S2052252520009653)

THE APS UPGRADE PROJECT – AN UPDATE

$815M project to update and renew the facility

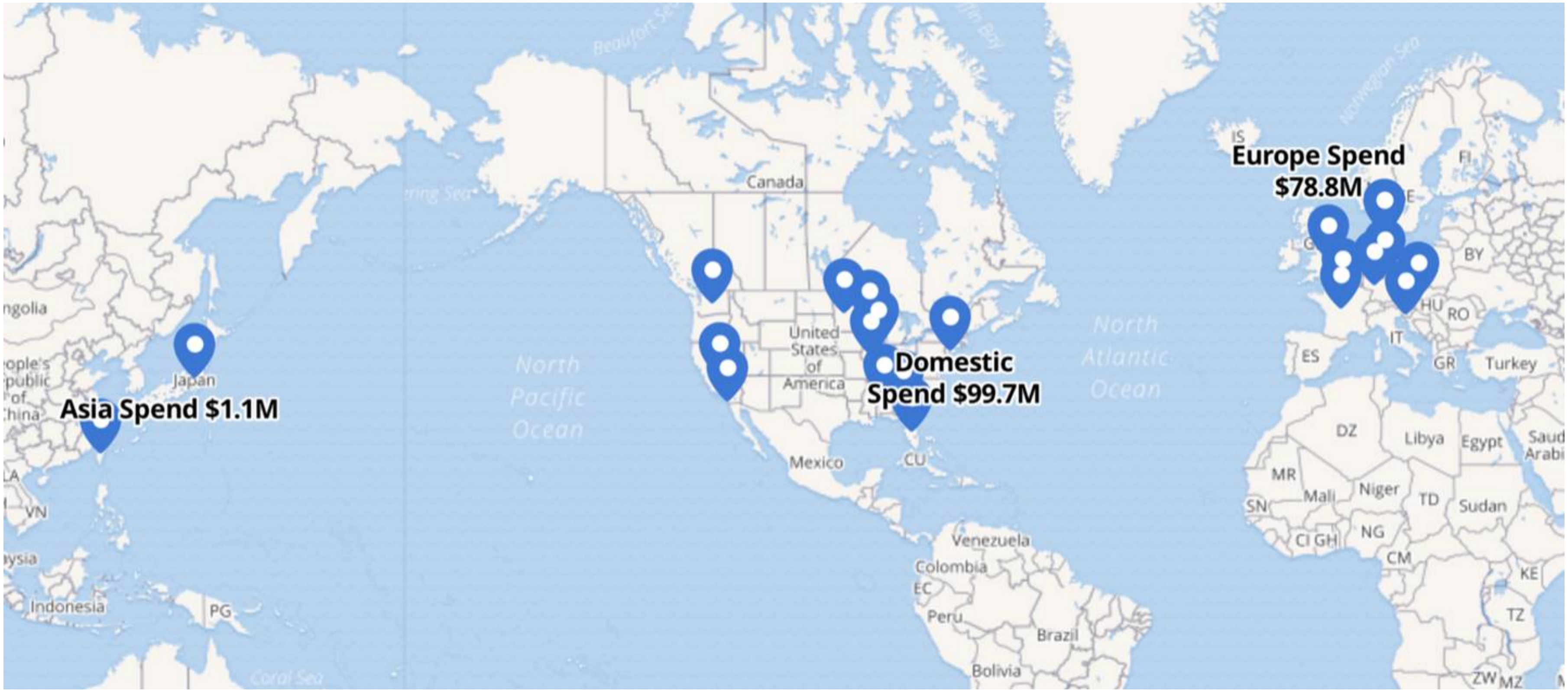
Re-uses $1.5B in existing APS infrastructure


15-ID (ChemMatCARS)
13-ID (GSECARS)
12-ID (XSD)
11-ID (XSD)
9-ID: CSSI (XSD)
Coherent Surface Scattering Imaging
8-ID: XPCS (XSD)
X-ray Photon Correlation Spectroscopy
7-ID (XSD)
6-ID (XSD)
5-ID (DND-CAT)
4-ID: POLAR (XSD)
Polarization Modulation Spectroscopy
3-ID (XSD)
2-ID (XSD)
1-ID (XSD)
APS Upgrade Project feature beamlines (Operator in parens)
APS Upgrade Project enhanced/improved beamlines (Operator in parens)
19-ID: ISN (XSD)
In Situ Nanoprobe
Long-Beamline Bldg.
20-ID: HEXM (XSD)
High-Energy X-ray Microscope
25-ID (XSD)
26-ID (CNM/XSD)
27-ID (XSD)
28-ID: CHEX (XSD)
Coherent High-Energy X-ray Sector for In Situ Science
30-ID (XSD)
32-ID (XSD)
33-ID: Ptycho (XSD)
PtychoProbe
34-ID: ATOMIC and 3DMN (XSD)
Atomic and 3D Micro & Nano Diffraction

- New storage ring
- 9 new feature beamlines + Long Beamline Building
- Enhancements for many other beamlines
- Combined, these improvements result in **brightness increases of up to 500x**

APS-U PROCUREMENT LOCATIONS

- Upgrade is ~2/3rd industrial procurements.
- Vendor oversight will be key in coming years


Europe Spend $78.8M
Domestic Spend $99.7M
Asia Spend $1.1M
Canada
United States of America
Mexico
Japan
North Pacific Ocean
North Atlantic Ocean
Venezuela
Colombia
Peru
Brazil
Bolivia
Indonesia
Turkey
Libya
Egypt
Sudan
Niger
Mali

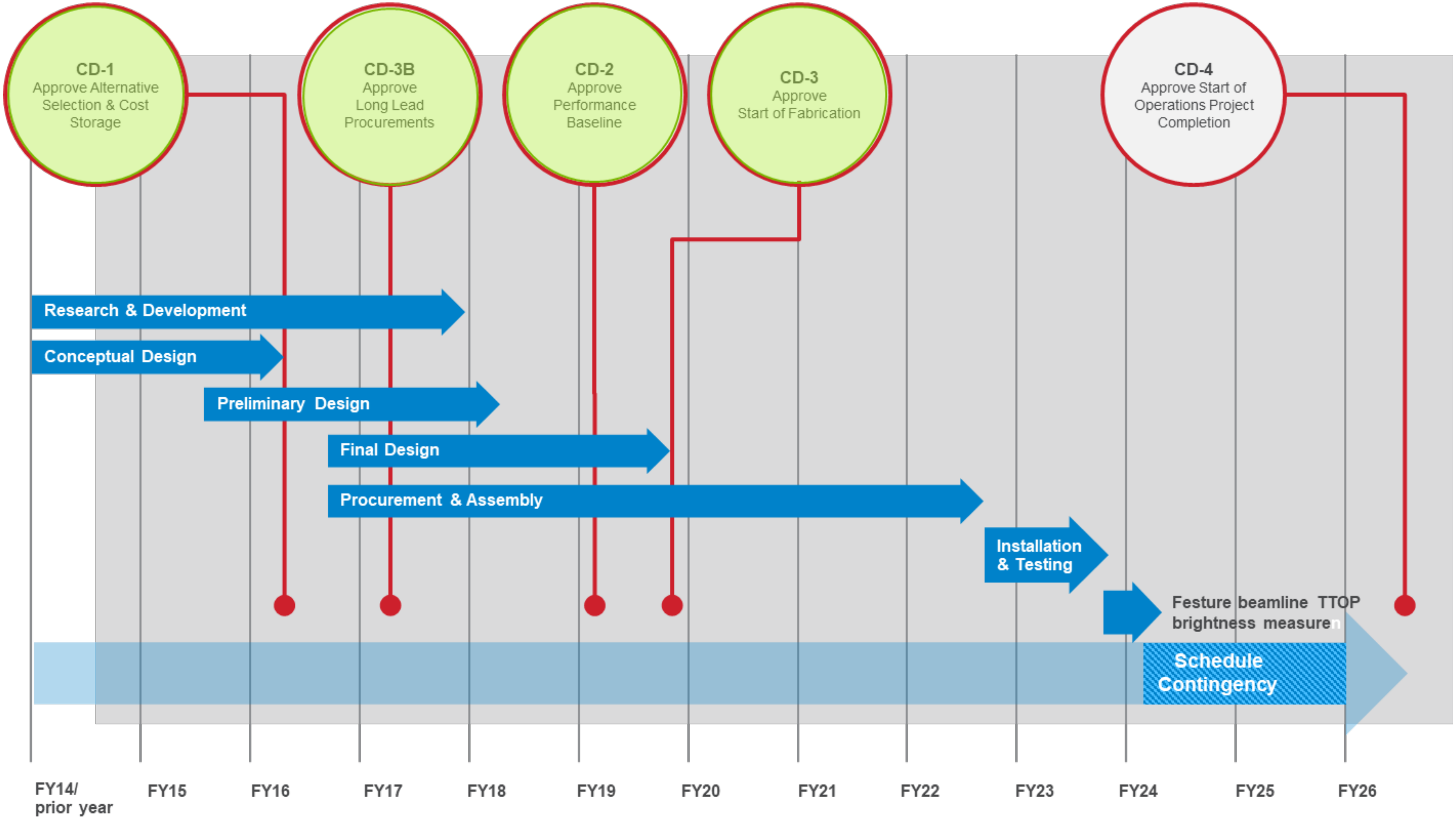
OFF-SITE BUILDING SOUTH OF ANL ON LEMONT ROAD

- 108,000 ft^2 - target beneficial occupancy Dec 2020 (we got it!)
- Maintenance and custodial contract in place
- Security in-place & in use
- Storage Racks installed

HILMAN
TK-EVO

APS-U SCHEDULE

TO DATE: NO CHANGES IN THE POSTED SCHEDULE



CD-1
Approve Alternative Selection & Cost Storage

CD-3B
Approve Long Lead Procurements

CD-2
Approve Performance Baseline

CD-3
Approve Start of Fabrication

CD-4
Approve Start of Operations Project Completion

Research & Development

Conceptual Design

Preliminary Design

Final Design

Procurement & Assembly

Installation & Testing

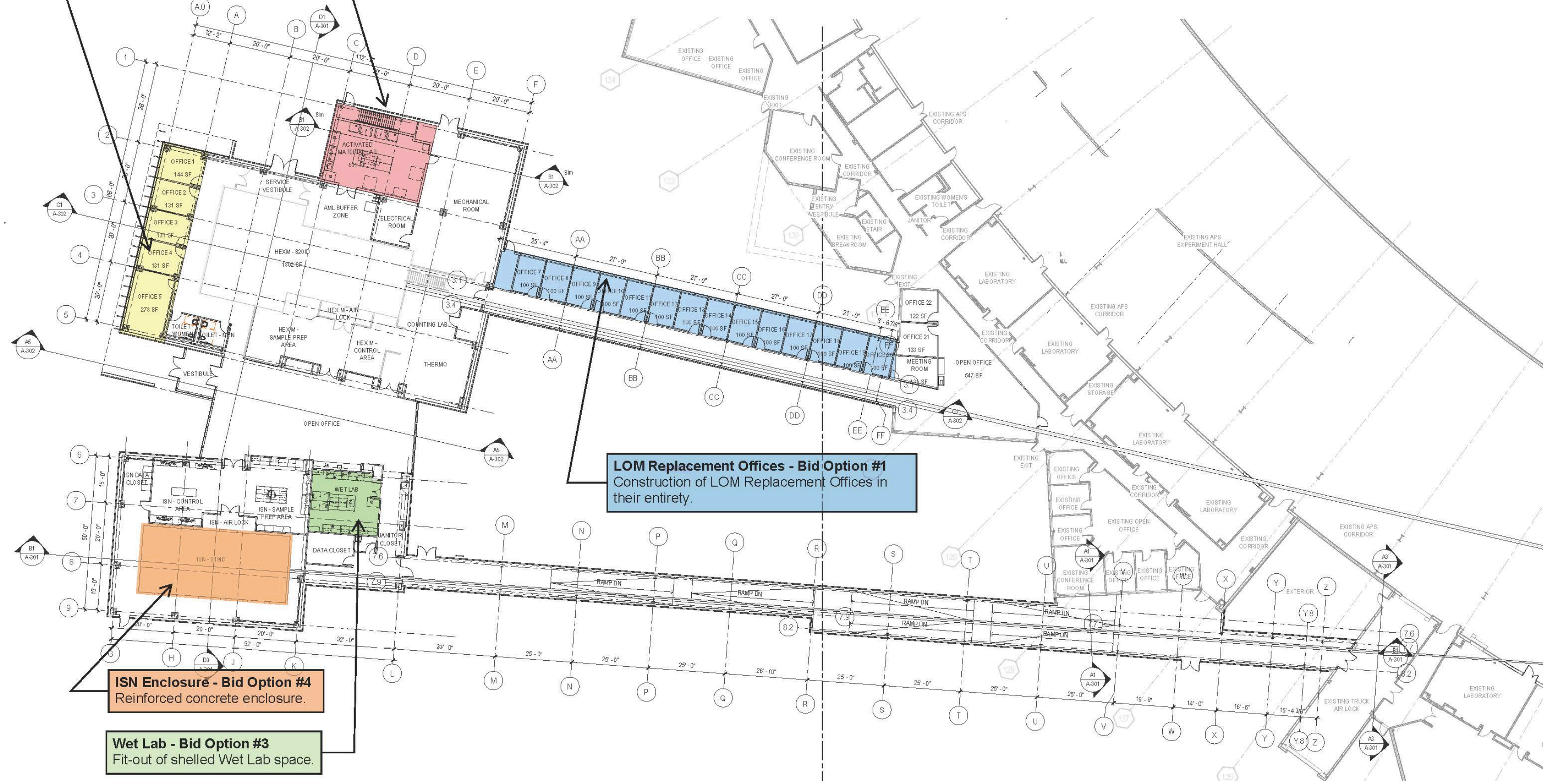
Festure beamline TTOP brightness measuren

Schedule Contingency

FY14/ prior year
FY15
FY16
FY17
FY18
FY19
FY20
FY21
FY22
FY23
FY24
FY25
FY26

THE LONG BEAMLINE BUILDING (LBB)

Beneficial occupancy	3/2022
Final occupancy	3/2022



HEXM Offices - Bid Option #5
Construction of HEXM Offices in their entirety.

Activated Materials Laboratory - Bid Option #2
Construction of AML in its entirety.

LOM Replacement Offices - Bid Option #1
Construction of LOM Replacement Offices in their entirety.

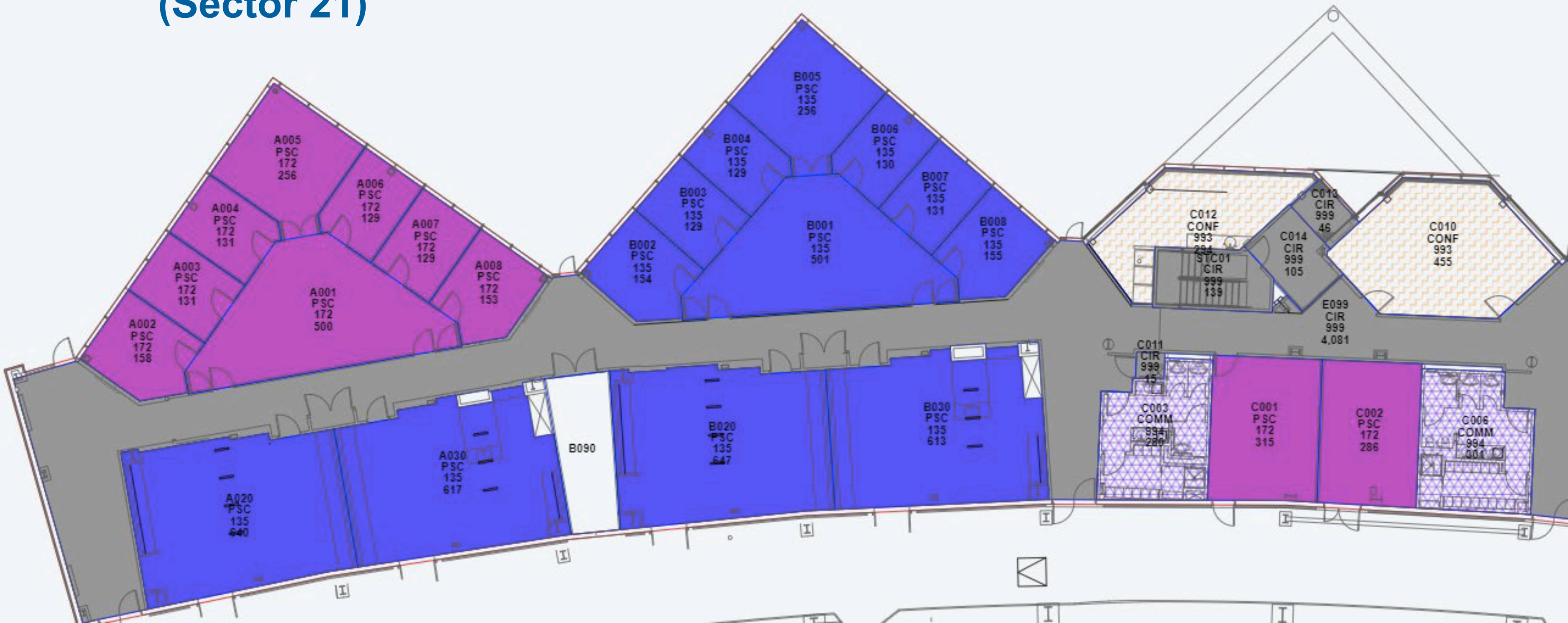
ISN Enclosure - Bid Option #4
Reinforced concrete enclosure.

Wet Lab - Bid Option #3
Fit-out of shelled Wet Lab space.

OFFICE 1
144 SF
OFFICE 2
131 SF
OFFICE 3
OFFICE 4
OFFICE 5
278 SF
ACTIVATED MATERIALS LAB
SERVICE VESTIBULE
AML BUFFER ZONE
ELECTRICAL ROOM
MECHANICAL ROOM
HEXM - S20ID
1302 SF
HEX M - AIR LOCK
HEX M - SAMPLE PREP AREA
HEX M - CONTROL AREA
COUNTING LAB
THERMO
TOILET - WOMEN
TOILET - MEN
VESTIBULE
OPEN OFFICE
OFFICE 7
100 SF
OFFICE 22
122 SF
OFFICE 21
133 SF
MEETING ROOM
OPEN OFFICE
547 SF
ISN DATA CLOSET
ISN - CONTROL AREA
ISN - AIR LOCK
ISN - SAMPLE PREP AREA
WET LAB
ISN - S19ID
DATA CLOSET
JANITOR CLOSET
RAMP DN
EXISTING OFFICE
EXISTING EXIT
EXISTING CONFERENCE ROOM
EXISTING CORRIDOR
EXISTING ENTRY VESTIBULE
EXISTING STAIR
EXISTING BREAKROOM
EXISTING APS CORRIDOR
EXISTING WOMEN'S TOILET
JANITOR
EXISTING LABORATORY
EXISTING APS EXPERIMENT HALL
EXISTING STORAGE
EXISTING OPEN OFFICE
EXISTING CONFERENCE ROOM
EXTERIOR
EXISTING TRUCK AIR LOCK

RELOCATION OF EXISTING OCCUPANTS

- Status
  - Build out of new office space has been completed on the Experimental Hall Floor
  - APS still owes LS-CAT a couple of more offices (most likely located in Sector 19 LOM)



SER-CAT
(Sector 21)

LS-CAT
(Sector 22)

A005 PSC 172 256
A006 PSC 172 129
A004 PSC 172 131
A007 PSC 172 129
A003 PSC 172 131
A001 PSC 172 500
A008 PSC 172 153
A002 PSC 172 158
B005 PSC 135 256
B004 PSC 135 129
B006 PSC 135 130
B003 PSC 135 129
B007 PSC 135 131
B001 PSC 135 501
B008 PSC 135 155
B002 PSC 135 154
C012 CONF 993 294
STC01 CIR 999 139
C013 CIR 999 46
C014 CIR 999 105
C010 CONF 993 455
E099 CIR 999 4,081
C011 CIR 999 15
C003 COMM 994 280
C001 PSC 172 315
C002 PSC 172 286
C006 COMM 994 301
A020 PSC 135 640
A030 PSC 135 617
B090
B020 PSC 135 647
B030 PSC 135 613

LBB STATUS

- Work (as of Dec 2020)
  - Installation of erosion controls is complete
  - Installation of construction fence is complete
  - Stripping of soil has begun.
  - Demolition of parking lot and sidewalks – via saw cutting to decrease vibrations
  - Ongoing vibration testing
  - Start of pot-holing investigative work
  - Construction of temporary parking lot in area of future parking lot

BAD
VIBRATIONS?

Questions?